EXPERIMENT :-5(A)

CODE:-

```
#include <stdio.h>

#define INF 9999

int main() {
   int n, i, j, source;
   int graph[20][20], dist[20], visited[20];
   int min, u;

   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter adjacency matrix:\n");
   for (i = 0; i < n; i++)
      for (j = 0; j < n; j++)
         scanf("%d", &graph[i][j]);

   printf("Enter source vertex: ");
   scanf("%d", &source);

   for (i = 0; i < n; i++) {
      dist[i] = graph[source][i];
      visited[i] = 0;
   }
```

```
dist[source] = 0;
visited[source] = 1;

for (i = 1; i < n; i++) {
  min = INF;
  u = -1;

  for (j = 0; j < n; j++) {
    if (!visited[j] && dist[j] < min) {
      min = dist[j];
      u = j;
    }
  }

  if (u == -1)
    break;

  visited[u] = 1;

  for (j = 0; j < n; j++) {
    if (!visited[j] &&
      graph[u][j] != 0 &&
      dist[u] + graph[u][j] < dist[j]) {
      dist[j] = dist[u] + graph[u][j];
    }
  }
}
```

```
    printf("\nShortest distances from vertex %d:\n", source);

    for (i = 0; i < n; i++)
        printf("To %d = %d\n", i, dist[i]);

    return 0;
}
```

Example

Graph: